
Graph-Based Screening for Potentially Cooperative Vessel Pairs: A Temporal Link Prediction Approach Using AIS (2012–2019)

Senior Project Report

Abstract

Illegal, unreported, and unregulated (IUU) fishing motivates monitoring **coordinated** vessel behavior. This project builds an **unsupervised** pipeline: **daily** graphs from AIS (2012–2019), a **Temporal Graph Convolutional Network (TGCN)** for **temporal link prediction**, and **independent geographic checks** on raw tracks to prioritize high-scoring pairs. Training edges use a 10 km same-day proximity rule; stricter distance screens (e.g., 25 km) are validation-only. There are no labels for “cooperative” vessels—the model learns from graph structure and time, and outputs candidates for review, not ground truth.

On the combined graph (proximity edges plus same-MMSI-prefix “social” edges, built with a corrected three-digit MID rule), a reproducible laptop run (1450 training time buckets, 3 epochs, seed 1) achieves **ROC AUC = 0.730** and **AP = 0.714** on 876 held-out daily buckets. A proximity-only full graph reaches higher scores (~ 0.78 AUC) under similar memory limits. Eight candidate pairs pass a strict 25 km, ± 1 day proximity filter; a gear-aware re-ranking shows that most excluded candidates are same-gear, same-flag trawler pairs whose co-location reflects normal fleet behavior. We do **not** claim legal findings or confirmed “dark fleet” membership.

Keywords: AIS, IUU fishing, temporal graph neural networks, link prediction, vessel co-movement

1 Introduction

Problem. Fisheries enforcement and research need ways to highlight vessels that may act together—transshipment, shared fishing grounds, or periods with missing AIS (“dark” behavior). The Automatic Identification System (AIS) provides large-scale movement data, but **cooperation is rarely labeled** in the data we use.

What this work does. We treat the problem as **temporal link prediction** on daily graphs: if the model assigns a high score to a vessel pair in a future time bucket, that pair is a candidate for analysts. We then validate candidates with simple proximity rules (distance between daily mean positions) and case studies (tracks, monthly overlap).

What this work does not do. We do not prove coordination or IUU activity; we provide a **ranked list plus evidence** that human review can interpret.

Contributions.

1. Reproducible pipeline: AIS $\rightarrow$ daily graphs $\rightarrow$ TGCN with temporal node features $\rightarrow$ time-based train/test split.

2. **MID-correct** combination of proximity and same-prefix social edges.
3. Reported full-coverage metrics under realistic laptop memory limits (train buckets capped where needed).
4. Validation and interpretation: proximity filters, eight-pair overlap plots, and a **gear-aware stratification** (§4.2.1) that uses fleet-wide co-location baselines to explain why most excluded candidates are normal fleet behavior.

Roadmap. §3 summarizes the link-prediction setup (full notation in Appendix C), then data, graph rules (10 km proximity + MID social edges), training objective, and post-hoc validation distances. §4 gives quantitative results (Table 2), geographic and interpretability figures (Tables 3–4, Figures 1–4), and Appendix A for gear definitions. §5–6 summarize limitations and conclusions; Appendix B lists reproducibility commands.

For readers and committees. (1) Training graphs use 10 km same-day proximity edges (plus optional MID “social” edges); 25 km and wider checks are post-hoc validation, not class labels. (2) The headline 0.730/0.714 scores refer to the combined full-coverage graph in Table 2; proximity-only and capped rows are different experiments—see §3.6. (3) Figure 2 (gear-aware stratification) uses 2012–2019 fleet data to show why most TGCN candidates are routine fleet co-locations; the 8 validated pairs survive because they are statistically unusual.

2 Background and Related Work

Fisheries monitoring and AIS. Illegal, unreported, and unregulated (IUU) fishing and at-sea coordination are active research areas. Large-scale studies use AIS broadcasts to map fishing effort and vessel encounters [Kroodsma et al., 2018]. AIS does not, by itself, label “cooperative” or illicit behavior; analysts typically combine movement data with proximity, registry, or enforcement context. That gap motivates unsupervised tools that rank pairs for review.

AIS-based fisheries and encounter analytics. Beyond footprint mapping, vessel-level AIS supports encounter and behavior analysis (e.g., transshipment and loitering [Miller et al., 2018]; monitoring-system assessments [Park et al., 2020]). That line of work motivates movement-based screening but often relies on rules or supervised targets when labels exist. Here we stay unsupervised at the learning stage and reserve geography for post-hoc validation.

Graph learning and link prediction. Co-presence is naturally a graph: vessels are nodes; edges encode same-day proximity (and optional same-registry structure). Graph convolutional encoders [Kipf and Welling, 2017] and inductive methods on large graphs [Hamilton et al., 2017] learn vector representations of nodes from topology and attributes. Link prediction scores missing or future links [Lü and Zhou, 2011]; temporal graph models (e.g., spatio-temporal convolutions [Yu et al., 2018]; recurrent or memory-based temporal graph networks [Rossi et al., 2020]) apply when G_t varies across t . Our implementation uses a TGCN-style recurrent update over daily snapshots with inner-product scores for pairs (see §3.4).

Evaluation philosophy. We use a chronological train/test split over daily buckets: train on earlier days, evaluate on later days. ROC AUC and AP therefore measure forecasting-style generalization to future co-occurrence, not i.i.d. mixing of time points.

3 Methods

3.1 Problem Formulation

Each calendar day (bucket) t defines a snapshot undirected graph $G_t = (V, E_t)$: nodes V are vessel MMSIs; edges E_t follow the construction rules in §3.3—primarily same-day positions within 10 km (Haversine), plus optional same three-digit MID social edges. The learning task is temporal link prediction: a TGCN produces node embeddings so that inner-product scores between pairs rank

observed edges above sampled non-edges under binary cross-entropy with logits, with no “cooperative” supervision—only structure and time. Train and test partition time buckets chronologically ($\sim 70\%/30\%$); ROC AUC and AP measure ranking quality on held-out future buckets.

Appendix C lists symbols, restates the setup for readers who prefer a notation table, and separates training-edge distance (10 km) from validation-only distances.

3.2 Data and Time Buckets

We use AIS-derived records spanning 2012–2019, aggregated to daily resolution ($\sim 2.38\text{B}$ vessel-day records; sea-surface temperature merged where available). The timeline is divided into daily buckets; 876 buckets enter the main edge list used for evaluation.

3.3 Graph Construction

- **Nodes** = MMSI identifiers (vessels).
- **Proximity edges**: same calendar day (daily time floor, typically 1D), pairwise great-circle (Haversine) distance at most 10 km between daily position summaries.
- **Social edges**: undirected links between vessels sharing the same three-digit ITU MID (national/organizational MMSI prefix). To keep graphs tractable, social edges are capped (e.g., up to 2000 per bucket).
- **Combined edge list** for the primary social-augmented benchmark.

Validation distances (not training edges). Post-hoc proximity checks for candidates use 25/50/100 km (and optional ± 1 day windows) on daily mean positions—different from the 10 km graph edge rule; §3.5.

3.4 Model and Training Objective

Architecture. Temporal Graph Convolutional Network (TGCN) with optional temporal node features (degree, interaction counts, partner diversity, recency, inter-event gaps).

Training details. Unless noted otherwise: embedding dimension 32, Adam optimizer, learning rate 0.001, 3 training epochs over ordered training buckets, binary cross-entropy with logits comparing positive edges to randomly sampled negative pairs (same count as positives per snapshot). Chronological split ratio $\sim 70/30$. Train-bucket count was capped at 1450 for the headline combined-graph run due to laptop RAM (higher caps caused OOM).

Evaluation. ROC AUC and AP on test buckets; edge deduplication rules avoid trivial leakage of training links into the test objective.

Computational cost. All experiments ran on a single laptop (Apple M-series, 16 GB unified RAM). The primary combined-graph run (1450 train buckets, 3 epochs) completed in roughly 20–40 minutes wall-clock; peak resident memory sat near 12–14 GB. No GPU was used; all matrix operations ran on CPU via PyTorch.

3.5 Heuristic Validation (Not Supervised Labels)

For candidate pairs we compute Haversine distance between daily mean positions and count days within 25/50/100 km (and ± 1 day alignment where noted). This is post-hoc geography, not a training label. A gear-aware stratification step (§4.2.1) further characterizes excluded candidates by comparing their gear-pair profiles against fleet-wide co-location baselines.

3.6 Which Numbers Are “The Same Experiment”?

Readers should keep the following settings separate (do not mix metrics across rows):

Table 1: Experimental settings and their roles.

Setting	Role
2012–2019 combined graph (prox. + social)	Primary benchmark —full timeline, social edges, reported AUC/AP on 876 test buckets.
Proximity-only graph	Ablation—no social edges; often higher AUC; shows contribution of the social layer.
Capped / smaller graphs	Sanity checks and tuning—easier prediction; very high AUC; not the same difficulty.
August 2017 short window	Exploratory only—few buckets; metrics are not comparable to the main table.

Table 2: Link-prediction results across settings (see §3.6 for comparability notes). The **combined + social** row is the primary benchmark.

Setting	Train cap / notes	Test eval	ROC AUC	AP
Proximity-only, full	~1500 train buckets	time split	0.776	0.795
Combined + social	1450 buckets, 3 ep, seed 1	876 test	0.730	0.714
Capped graph	tuning / easier task	—	~0.95	~0.96
Capped (5-fold CV)	rolling CV	—	0.927±0.067	0.947±0.048
Aug 2017 (case-study)	small	5 test buckets	~0.56	~0.67

4 Results

§4 presents quantitative link-prediction metrics (Table 2), followed by geographic validation and interpretability figures. Recall from §3.6 that table rows represent different experimental settings—do not mix metrics across them.

4.1 Quantitative Performance (TGCN)

Primary run details (combined row). Embedding dimension 32, learning rate 0.001, temporal node features enabled. Increasing allowed training buckets (e.g., 400 → 1450) improved AUC/AP until memory limits on the laptop (~1480–1500 buckets caused OOM/killed).

Variability and uncertainty. The headline combined run is a single-seed report (seed = 1), so cross-seed confidence intervals are not available for the primary row. Two other sources give a sense of variability:

- **Per-bucket dispersion.** Across the 876 test buckets the per-bucket standard deviation is ± 0.103 for ROC AUC and ± 0.106 for AP.
- **Cross-seed reference (capped graph).** On the easier capped graph, a 5-fold rolling-window CV with seeds 1–5 yields 0.927 ± 0.067 AUC and 0.947 ± 0.048 AP.

Why do social edges lower AUC? The proximity-only graph scores ~0.78 AUC versus ~0.73 for the combined graph. Three factors likely contribute:

1. **Noise from registry grouping.** MID social edges link every pair of vessels sharing the same three-digit national prefix. In a predominantly Chinese-flagged dataset (MID 412), this connects thousands of vessels that never physically co-occur.
2. **Over-smoothing.** Adding dense social edges increases the effective neighborhood size for each GCN layer, blurring node embeddings toward a common mean.
3. **Objective dilution.** The training loss treats social edges and proximity edges identically (both positive examples). The model spends capacity fitting registry structure rather than geographic co-occurrence.

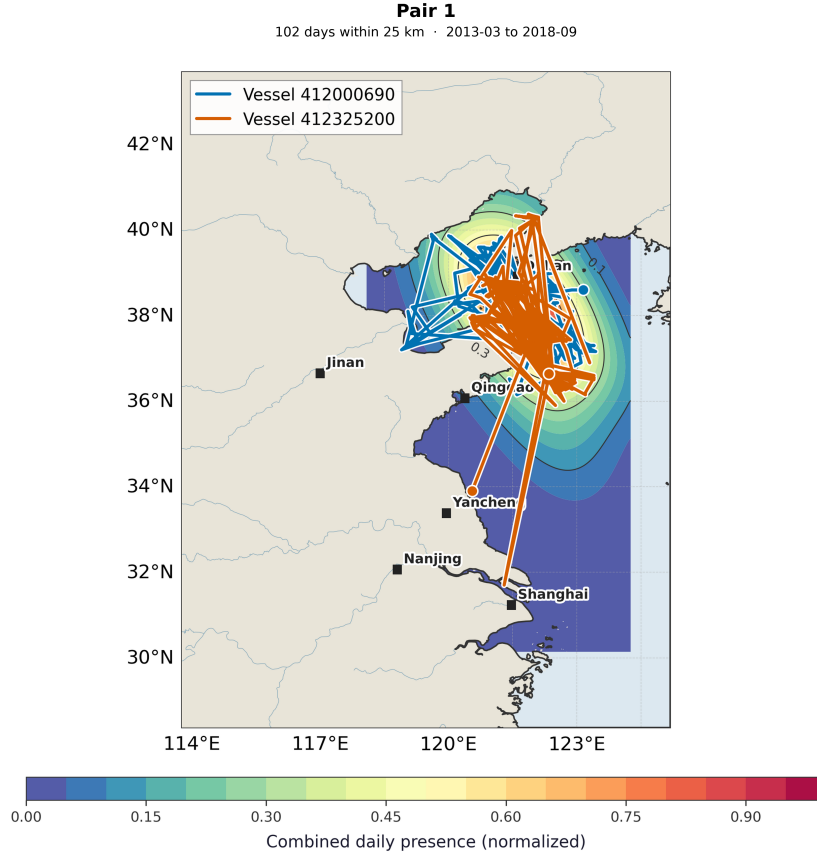


Figure 1: **Pair 1**—vessels 412000690 (blue) and 412325200 (orange), Chinese coast, 2013–2018, 102 days within 25 km. Background = combined daily presence density (KDE, 0–1 scale). The model’s top-ranked pair shows sustained co-presence in the Yellow Sea ($\sim 121^\circ\text{E}$, $37\text{--}39^\circ\text{N}$), confirming the high link-prediction score corresponds to a real shared operating area.

4.2 Geographic Validation and Case Studies

Among workflows that rank and check top candidates, **eight** pairs pass a 25 km, ± 1 day rule. Full monthly overlap counts (Figure 3) range from 84 days (Pair 1) down to 4 days (Pair 8). Figure 1 showcases vessel pair 412000690 $\leftrightarrow$ 412325200 (102 days within 25 km), drawn from a separate case-study list. §4.2.1 examines the gear-type composition of all 200 candidates and explains why the other 192 are consistent with normal fleet behavior.

4.2.1 Gear-Aware Candidate Stratification

The TGCN ranks roughly 200 candidate pairs, yet only 8 survive the 25 km / ± 1 day geographic filter. What explains the other 192? A gear-aware re-ranking pipeline sheds light on the composition of excluded candidates and provides a quantitative justification for the filtering.

Method. Each of the 267 unique MMSIs in the top-200 list was enriched with a dominant gear type by sampling 96 days of MMSI daily positions (one per month, 2012–2019) and joining them to cell-aggregated fleet daily data. Separately, a gear-pair co-location baseline was computed from 32 sampled fleet days across 2012–2019: for each grid cell-day, we recorded which gear types were present and tallied co-occurrence counts across ~ 7 million cell-days. This baseline quantifies how often each gear combination shares the same location by default—regardless of any model score.

Each candidate pair then received a gear-adjusted score:

$$S_{\text{adj}} = S_{\text{orig}} \times \frac{1}{1 + \alpha \cdot \hat{B}} \quad (1)$$

where $\hat{B}$ is the normalized baseline rate for the pair’s gear combination and $\alpha = 2$. Pairs whose gear combination has a high fleet-wide co-location rate receive a larger discount; cross-gear or rare-gear pairs are less affected.

Key findings (Figure 2).

- **75% of the top-200 candidates** are trawler-dominated: 82 pairs (41%) are Trawlers + Trawlers and 67 (34%) are Line/generic + Trawlers. However the two categories tell very different stories: Trawlers + Trawlers is actually $1.1\times$ *below* the fleet baseline, while Line/generic + Trawlers appears $10.5\times$ *above* the fleet baseline—the TGCN is specifically surfacing these cross-gear encounters far more than chance would predict.
- **50% are same-flag** (predominantly CHN + CHN), consistent with vessels from the same national fleet operating in the same waters.
- After gear adjustment, cross-gear validated pairs held or improved their rank: the Set gill-nets + Trawlers pair jumped 48 ranks ($72 \rightarrow 24$), and the Fixed gear + Trawlers pair dropped only 2 ranks.
- The two Trawlers + Trawlers validated pairs dropped 70+ ranks each (e.g., $80 \rightarrow 154$), confirming their co-presence is explained by shared fishing grounds.

Why the results look this way. The top-200 list is dominated by trawler pairs for a straightforward reason: trawlers are the most common vessel type in the dataset and they concentrate in the same high-density fishing grounds (Yellow Sea, East China Sea). From the TGCN’s perspective, two trawlers that share the same seasonal migration look structurally identical—they appear in the same grid cells, on the same days, year after year. The model correctly identifies this overlap but cannot distinguish “two boats following the same fish” from “two boats deliberately meeting.” That is why 80 of the 82 trawler–trawler candidates fail the strict 25 km / ± 1 day check: their graph-level similarity does not translate to sustained physical proximity on a given day.

Cross-gear pairs tell the opposite story. A fixed-gear vessel (stationary nets or traps) and a trawler (moving nets) have different operating modes, target different species, and rarely share the same grid cell by chance. When the model flags such a pair *and* the geographic filter confirms they were repeatedly within 25 km, the co-location is unlikely to be coincidental. This is reflected in the pass rates: Fixed gear + Trawlers passes at 25% (1 of 4), compared to 2.4% for trawler–trawler pairs—roughly a $10\times$ difference. The intermediate categories follow the same gradient: Set gillnets + Trawlers passes at 8.3%, and Line/generic + Trawlers at 6.0%.

What this means for the project.

1. **It justifies the geographic filter.** The 192 excluded pairs are not “false positives” in the traditional sense—the model genuinely detected shared movement patterns. But those patterns are explained by normal fleet behavior, not by vessel-to-vessel coordination. The filter correctly separates fleet-level noise from pair-specific signal.
2. **It strengthens the 8 validated pairs.** Cross-gear pairs survive at disproportionately high rates, meaning the validated set is enriched for behaviorally unusual encounters—exactly the kind of co-location that warrants further investigation.

4.3 Monthly Overlap and Cluster Context

Figure 3 charts the monthly count of days each validated pair’s vessels were within 25 km. Figure 4 maps those same rendezvous locations geographically and highlights a six-vessel subgroup near $\sim 30^\circ\text{N}$, 122°E .



Gear from cell-aggregated fleet daily data (2012–2019).

Figure 2: (a) All 200 top candidates by gear combination: stacked counts of pairs that failed vs. passed the 25 km / ± 1 day geographic check. (b) The 8 validated pairs with gear type, flag, proximity days, and TGCN rank.

4.3.1 Gear Type Codes

Legend labels on Figures 3–4 and Tables 3–4 show **country** · **gear** per vessel. Country comes from the cell-fleet enrichment when the MMSI appears there, otherwise from the ITU MID prefix. Gear strings (e.g., trawlers, set_gillnets, fixed_gear, fishing) are geartype labels from cell-aggregated fleet files, assigned by hours-weighted overlap between MMSI daily cells and fleet rows. They indicate broad fisheries-activity categories, not a formal gear license per vessel. Figures and tables use the same 2012–2019 enrichment. Plain-language definitions are in Appendix A.

4.4 Exploratory Short-Window Analysis (August 2017)

An early TGCN run on a narrow August 2017 slice (five buckets, ~ 0.56 AUC, ~ 0.67 AP) was used for exploratory interpretability; its metrics are not comparable to the 2012–2019 benchmark (see §3.6). The gear-aware stratification in §4.2.1 supersedes that window by enriching all 200 candidates with full 2012–2019 fleet data.

Table 3: Eight validated pairs (country · gear per vessel).

src	dst	Country · gear (src dst)
412422375	412428225	CHN · fixed_gear CHN · trawlers
412461376	412415321	CHN · trawlers CHN · set_gillnets
412437423	412435485	CHN · trawlers CHN · fishing
412410128	412416248	CHN · fishing CHN · trawlers
412420679	412413383	CHN · fishing CHN · trawlers
412450427	111203412	CHN · fishing MID111 · trawlers
412985698	412443375	CHN · trawlers CHN · trawlers
412461376	412427825	CHN · trawlers CHN · trawlers

Table 4: Six-vessel cluster (Figure 4).

MMSI	Country · gear
412413383	CHN · trawlers
412420679	CHN · fishing
412427825	CHN · trawlers
412435485	CHN · fishing
412437423	CHN · trawlers
412461376	CHN · trawlers

5 Limitations

5.1 Threats to Validity

- **Link prediction is not a verdict.** ROC AUC / AP measure ranking of observed vs. sampled non-edges under the graph definition. A model can score well when vessels share fishing grounds, ports, or traffic lanes—behavior that is legal and common.
- **AIS selection and compliance.** Vessels may disable AIS, use identity changes, or have sparse coverage; the graph only sees reported positions.
- **Confounding by density.** Crowded regions produce many possible edges; degree and popularity effects can inflate link-prediction metrics.
- **Metric–semantics gap.** Good calibration of the learning objective does not imply validated enforcement outcomes.

5.2 Design and Data Constraints

- No behavioral ground truth for “cooperation” or IUU.
- Grid and sampling choices affect positions and edges.
- Laptop memory caps training buckets (1450 stable here; higher values failed).
- Shared geography vs. coordination: overlap in dense regions is ambiguous (§5.1).
- Country and gear on plots come from cell-aggregated fleet tables (§4.3.1, Appendix A)—descriptive context, not a formal registry. Gear labels support the gear-aware stratification in §4.2.1, but should not be treated as definitive vessel classifications.
- Gear-aware re-ranking (§4.2.1) uses a sampled baseline (32 fleet days across 2012–2019, ~7M cell-days). The discount formula (Eq. 1) is a heuristic, not a calibrated statistical model.
- **Single-seed headline numbers.** The primary combined-graph AUC/AP is from one seed; per-bucket dispersion and cross-seed references are discussed in §4.1, but full multi-seed CIs remain future work.

6 Conclusion and Future Work

We presented an unsupervised AIS → graph → TGCN pipeline with a time-split evaluation and an explicit link-prediction formulation (§3.1; Appendix C). The combined proximity + MID-correct

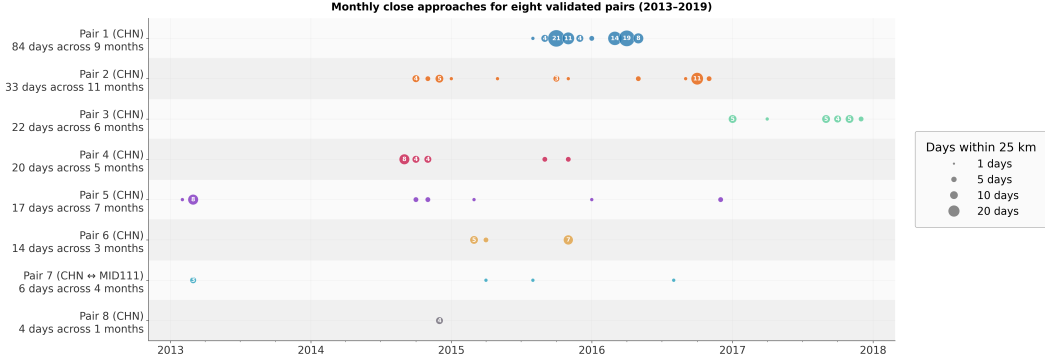


Figure 3: Bubble chart of monthly close approaches (days within 25 km) for all eight validated pairs, 2013–2019. Each row is one pair (sorted by total overlap, highest at top); bubble size $\propto$ days within 25 km that month. Pair 1 (CHN, 84 days across 9 months) dominates with large, closely spaced bubbles in 2015–2016, indicating persistent co-presence. Most other pairs are sporadic. Pair 7 is the only cross-flag pair.

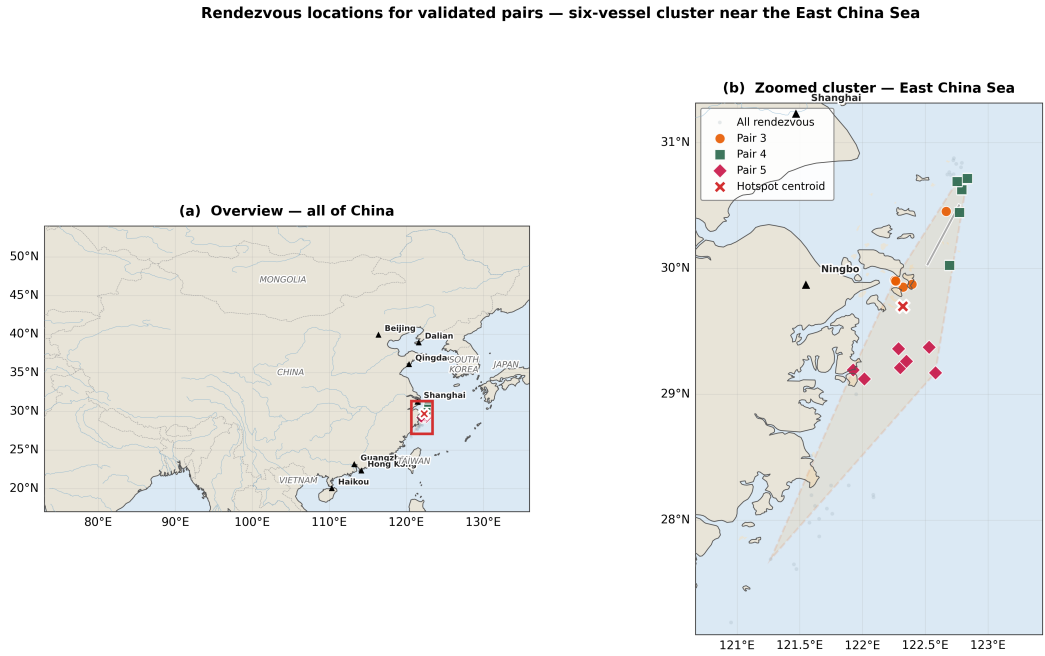


Figure 4: (a) Overview map of China showing the cluster region (red rectangle). (b) Zoomed view of the East China Sea cluster. Faint gray dots show every recorded rendezvous point; colored markers highlight three cluster pairs (Pair 3, 4, 5—matching Figure 3 numbering) involving six CHN-flagged vessels. The red X marks the weighted hotspot centroid ($\sim 29.7^\circ\text{N}$, 122.3°E), roughly 50–80 km offshore from Ningbo. Three independent pairs (six vessels) converge repeatedly on the same offshore area, suggesting a shared operating region rather than coincidence.

social graph yields **0.730 ROC AUC** and **0.714 AP** on 876 test buckets under the documented laptop protocol (Table 2), alongside proximity-only and capped-graph comparisons. Eight pairs pass strict post-hoc proximity screening (Table 3); a gear-aware stratification (§4.2.1) shows that 75% of top candidates are trawler-dominated same-fleet pairs whose co-location is consistent with normal fishing behavior, while the validated cross-gear pairs maintain or improve their ranking after adjustment. Interpretability figures add country and gear context where data allow (Appendix A). For practice, the output is a ranked candidate list plus reproducible geographic and fleet-context checks—not a determination of wrongdoing. We do not claim confirmed coordination or IUU.

Future work includes finer spatial/temporal resolution; owner or fleet metadata from registries; supervised or semi-supervised learning when reliable labels exist; tighter integration of movement-based gear or behavior classifiers; full-coverage multi-seed runs and larger training-bucket budgets on high-memory hardware; heterogeneous-edge or edge-weighted architectures to better exploit social edges without the over-smoothing and noise effects described in §4.1; and continued care in separating benchmark metrics (2012–2019) from illustrative short windows.

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A Gear Type Codes

A.1 Source of Labels

In this project, gear on figures and tables comes from cell-level fleet aggregates, not from a per-vessel registry. For each MMSI–day, we join AIS daily cells to fleet files where each row is hours by flag $\times$ geartype in that cell; the vessel is labeled with the geartype that receives the most weighted hours that day. The label is a coarse, behavioral proxy for “what kind of fishing that cell’s traffic was attributed to,” not a formal gear certificate.

A.2 Codes Used

A.3 How to Interpret Labels

1. Same MMSI, different days can get different geartypes if the vessel moves between cells dominated by different fleet attributions.
2. Same geartype on two MMSIs does not prove they use identical gear—only that both were attributed the same coarse class in the fleet enrichment window (2012–2019).
3. AIS “ship type” codes differ from fleet geartype here: the latter is fisheries-activity oriented, not the full IMO/AIS ship-type list.

Table 5: Gear type codes from the fleet geartype column.

Code	Short gloss
fishing	Generic/unspecified fishing. Catch-all category when activity is attributed to “fishing” without a more specific gear class.
trawlers	Trawling. Vessels tow nets through the water column or along the bottom (e.g., otter trawl, beam trawl).
fixed_gear	Stationary gear fixed to the seabed or structure: pots, traps, stakes, weirs, and similar non-towed gear.
set_gillnets	Set gillnets. Nets set and left (anchored, on bottom, or sometimes drifting) so fish gill in the mesh.
set_longlines	Set longlines. A long line with many baited hooks, deployed and left to soak.
other_purse_seines	Purse seining (other). A surrounding net used on schooling fish; the bottom is pulled closed.
pole_and_line	Pole-and-line / baitboat. Fish caught with hand-held poles and hooks, often with live bait.

Table 6: Reproducibility commands. All commands assume the project root directory. Long lines break automatically.

Item	Command / path
Combined edges	<code>python3 scripts/add_social_edges.py --edges artifacts/edges_2012_2019_full.parquet --out artifacts/edges_full_with_social.parquet --max-social-per-bucket 2000</code>
Primary TGCN	<code>artifacts/tgcn_social_maxb1450_ep3.json</code>
Case study (Fig. 1)	<code>python3 scripts/compute_pair_overlap_from_daily.py --contour</code>
Bubble chart (Fig. 3)	<code>PYTHONPATH=scripts python3 scripts/plot_pair_overlap_time_series.py</code>
Cluster map (Fig. 4)	<code>PYTHONPATH=scripts python3 scripts/analyze_six_vessel_cluster.py</code>
Gear re-ranking (Fig. 2)	<code>PYTHONPATH=scripts python3 scripts/gear_aware_reranking.py</code>

B Reproducibility

Eight-pair overlap (copy/paste from project root):

```
PYTHONPATH=scripts python3 scripts/overlap_by_month_8pairs.py \
--pairs artifacts/close_pairs_fullcoverage_25km_w1.csv \
--daily-root "data/MMSI daily vessels " \
--all-files --full-months \
--distance-km 25 --day-window 1 \
--out-csv artifacts/eight_pairs_overlap_by_month.csv \
--out-plot artifacts/plots/eight_pairs_overlap_by_month.png \
--enrichment artifacts/cooperative_pairs_with_flag_gear.csv
```

C Notation and Formal Problem Setup

C.1 Symbol Glossary

C.2 Graph Semantics

For each $t \in \mathcal{T}$, $G_t = (V, E_t)$ is undirected: $\{u, v\} \in E_t$ means the pair satisfies the edge construction rule for that day—spatial proximity within the 10 km same-day threshold (and optional same three-digit ITU MID links), not “ground-truth cooperation.” Node set V may be defined globally from all MMSIs in the study period or induced per implementation; edges are sparse and may be capped per bucket for memory (§3.3).

Table 7: Symbol table.

Symbol	Meaning
t	A calendar day (time bucket) in the AIS timeline.
$\mathcal{T}$	The set of days used after filtering (e.g., days with edges).
V	Set of nodes (9-digit vessel MMSIs).
$G_t = (V, E_t)$	Undirected snapshot graph for day t .
E_t	Edges for day t : proximity (§3.3) $\pm$ optional social edges.
u, v	Distinct nodes (unordered pair $\{u, v\}$).
$s_{u,v}$	Scalar score (logit) for pair (u, v) on a given forward pass / bucket.
$\mathbf{h}_u$	Embedding vector for node u after the TGCN (dimension 32 in the primary run).
$\mathcal{T}_{\text{train}}, \mathcal{T}_{\text{test}}$	Chronological partition of buckets ($\sim 70\% / 30\%$).

C.3 Temporal Link Prediction Objective (Informal)

The model maps the sequence $\{G_t\}$ (and optional temporal node features) to a recurrent hidden state and node embeddings. For each training snapshot, positive pairs are edges in $E_t \cap \mathcal{T}_{\text{train}}$; negative pairs are sampled from node pairs that are not positive for that snapshot. Scores take the form:

$$s_{u,v} \propto \mathbf{h}_u^\top \mathbf{h}_v \quad (2)$$

optimized with binary cross-entropy with logits (BCEWithLogitsLoss) against labels $y \in \{0, 1\}$. There is no label for IUU, transshipment, or “cooperative”—only structural positives and random negatives.

Evaluation. On $\mathcal{T}_{\text{test}}$, the model ranks candidate pairs; ROC AUC and Average Precision (AP) summarize how well observed test edges rank above non-edges. High scores mean plausible future co-occurrence under the graph definition, not culpability.

C.4 Training vs. Validation Distances

Table 8: Separation of training and validation distance thresholds.

Stage	Typical distance / rule	Role
Graph edges (training signal)	≤ 10 km, same day	Defines E_t for learning.
Candidate screening (post-hoc)	25 / 50 / 100 km, optional ± 1 day	Not used as training labels; validates high-scoring pairs on raw tracks (§3.5).